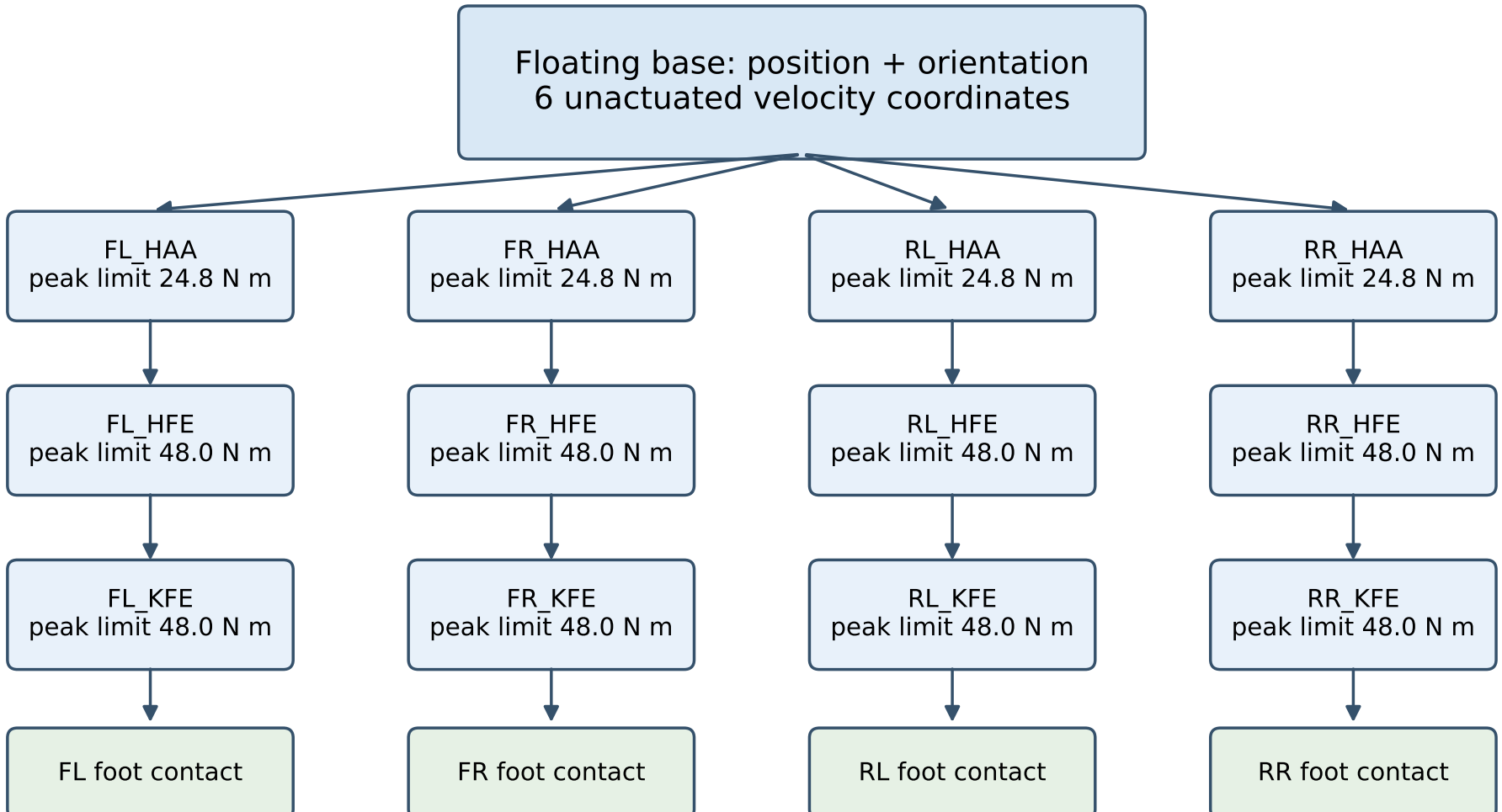
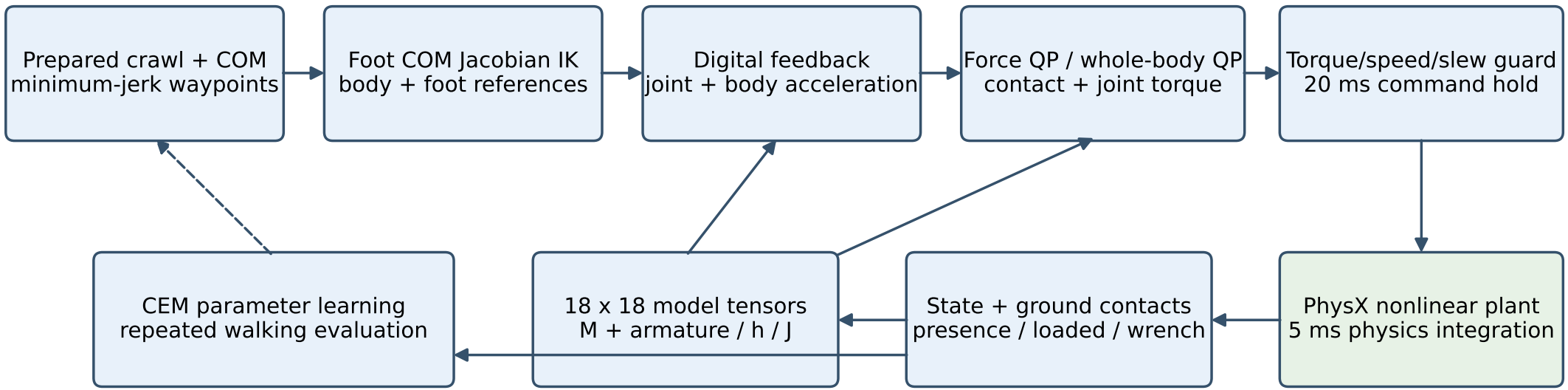
V16 | 12 actuated joints + a floating base



54 source rigid members: mass / COM / inertia aggregated without dropping fixed members.
M(Q) vdot + h(Q,v) = S^T tau + Jc(Q)^T lambda. v: 18 dimensions; torque input: 12.

V16 | digital model-based control and parameter learning



GUIDELINE: physical laws -> ODE -> Laplace + initial conditions -> latched step
-> continuous response -> sampled coupled difference -> B operator -> MIMO -> SISO channels -> recurrence

12 torque inputs; no virtual base actuator. Force-only QP fixes acceleration; whole-body QP can adjust it.
Both log finite-iteration constraint violations. COM reference bounds do not bound actual robot motion.
Safety projection can change torque within a hold interval. Linear/SISO derivation is local, not a global walking proof.
Learning tunes controller parameters; an independent seed evaluation is required before claiming success.